AP PRECALCULUS

Unit 1 • Lesson 1.7 • Guided notes

Topic 1.7 • Rational Functions and End Behavior

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Determine the end behavior of a rational function by comparing the degrees of numerator and denominator.
- Find horizontal and slant asymptotes, and write end behavior with limits.

Vocabulary. rational function, horizontal asymptote, slant asymptote, quotient of leading terms.

1. Quotient of leading terms

For $r(x) = \dfrac{p(x)}{q(x)}$, as $x \to \pm\infty$ the function behaves like _____.

Key idea

$\deg p < \deg q$: horizontal asymptote _____

$\deg p = \deg q$: horizontal asymptote $y =$ _____

$\deg p = \deg q + 1$: _____ asymptote, found by polynomial division

$\deg p > \deg q + 1$: behaves like a _____ of degree $\deg p - \deg q$

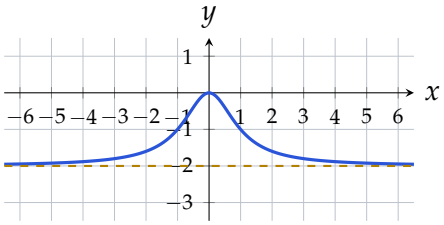
Example 1 • Limits at infinity

Find $\lim_{x\to\infty}$ for $\dfrac{3x+1}{x^2-4}$, $\dfrac{8x^2-x}{2x^2+5}$, $\dfrac{x^3}{x^2+1}$.

Example 2 • Slant asymptote

Find the slant asymptote of $s(x) = \dfrac{x^2+x-1}{x-1}$.

Example 3 • From a graph



The graph of $r(x) = \frac{-2x^2}{x^2+1}$ is shown. Write the end behavior with limits.

Common mistakes

- Reading the asymptote from the constant terms.
- Believing a graph can never cross a horizontal asymptote: it can, for moderate x.

Check your understanding

1. Horizontal asymptote of $\dfrac{5-2x^3}{x^3+x}$: _____.
2. Slant asymptote of $\dfrac{2x^2+3x}{x+1}$: _____.